Operation Guide for Configuration Modification and Other Operations of Xiaozhi AI Terminal Open Source Project (Part 1)

This document is intended only for developers, at least beginners who have already gotten started

Modification Premise:

1) Please first ensure that the ESP-IDF Development Environment has been installed and configured, with a version above IDF 5.0 (each major version of IDF is incompatible). To avoid unnecessary issues, it is recommended to use **IDF V5.3.2**.

For the installation and configuration of the ESP-IDF Development Environment on Windows 10, please refer to the previous tutorial document: Setting up the ESP IDF 5.3 Development Environment on Windows and Compiling Xiaozhi

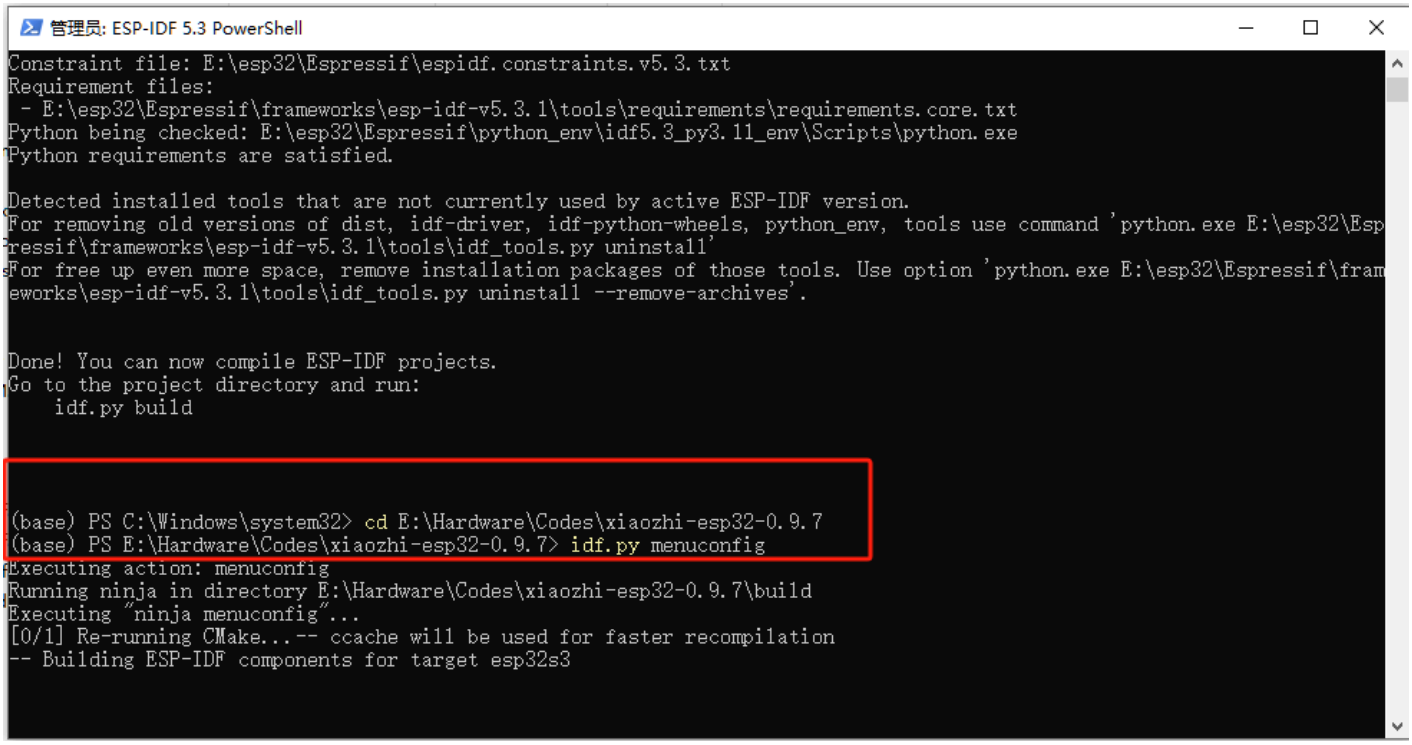
1. Open the operation of the Xiaozhi AI Chatbot Open Source Project

1) Use the PowerShell command terminal to switch to the code path of the Xiaozhi AI Terminal open-source project

Open the integrated ESP IDF 5.3 PowerShell command terminal, and use the cd command to switch the directory to the root directory of xiaozhi-esp32-0.9.7

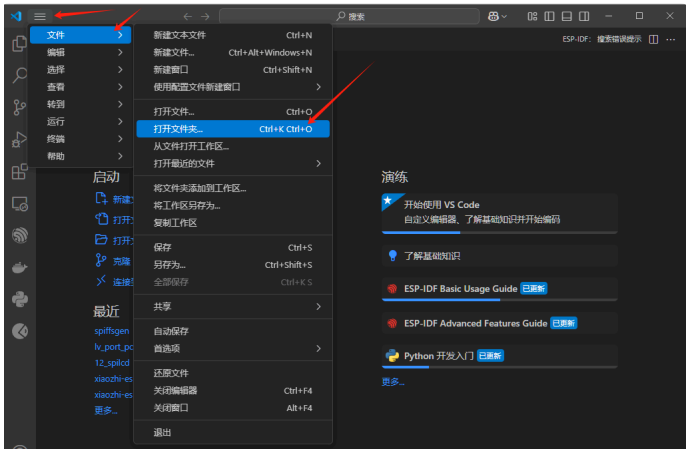
Note: The above 0.9.7 is the version number and the file directory path where the code is stored, provided only as an example for reference. Please modify it according to your own source code path and version number;

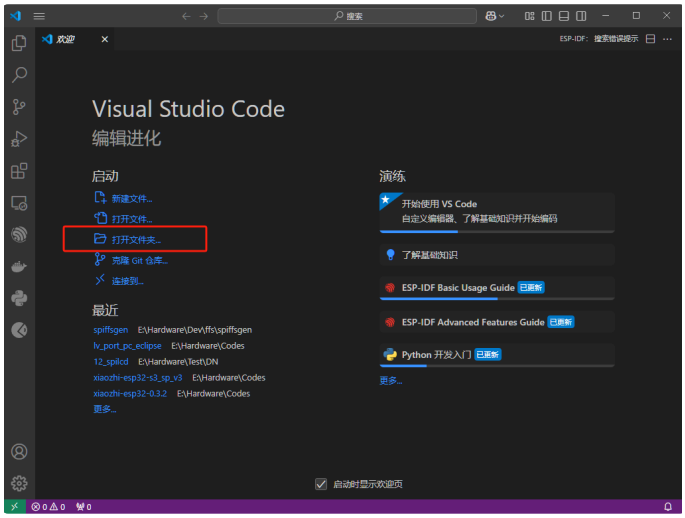
ESP-IDF 5.3 PowerShell is recommended to be opened in administrative mode.



2) Open the Xiaozhi AI Terminal open-source project using VSCode

① Open VScode, select "Open Folder" on the start page (right figure below) or in the menu - File - Open Folder (left figure below)

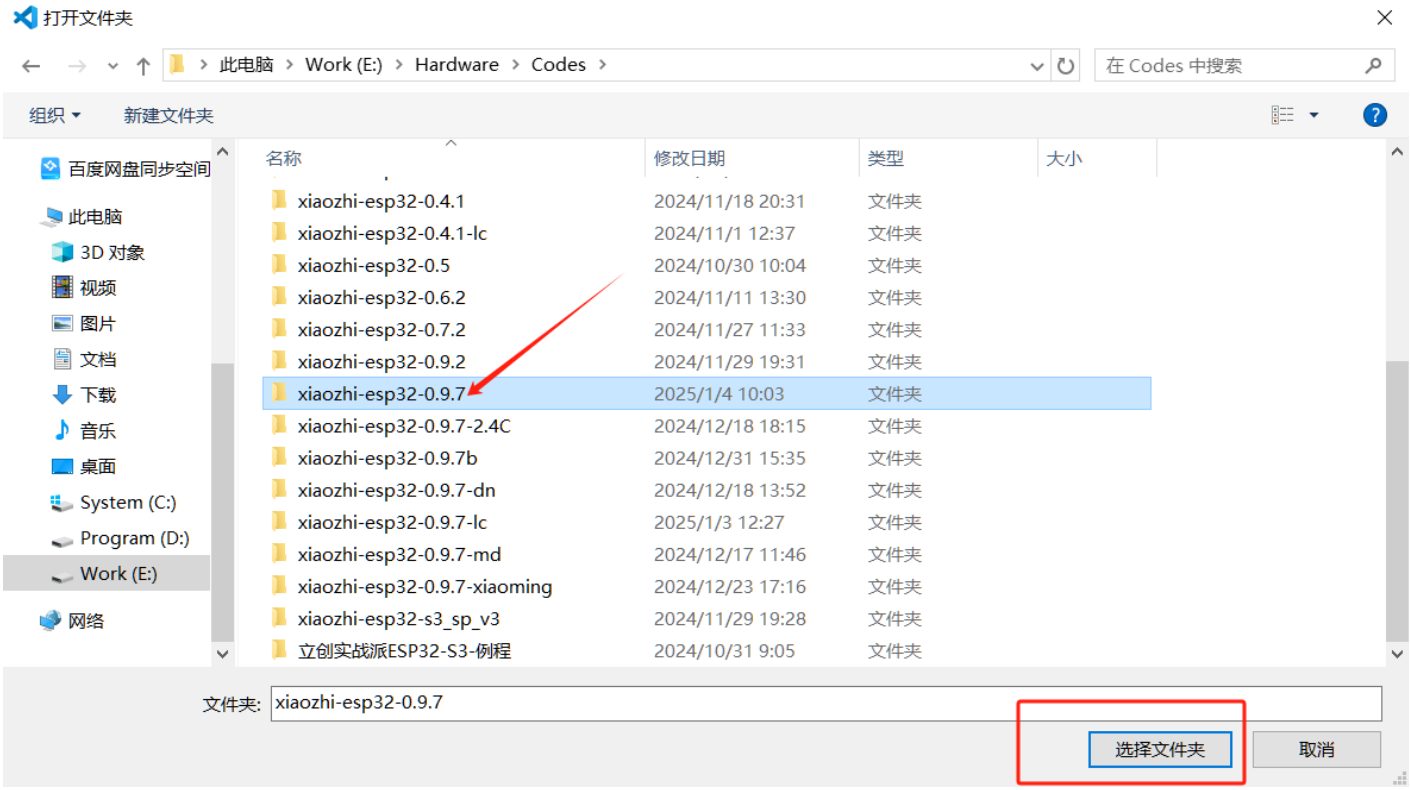




② Open the directory to the decompressed Xiaozhi AI source code stored on your computer and directory

Please choose the folder path opened by VScode to the root directory of the Xiaozhi AI source code by yourself**, and do not choose to enter the main directory under the project root directory.**

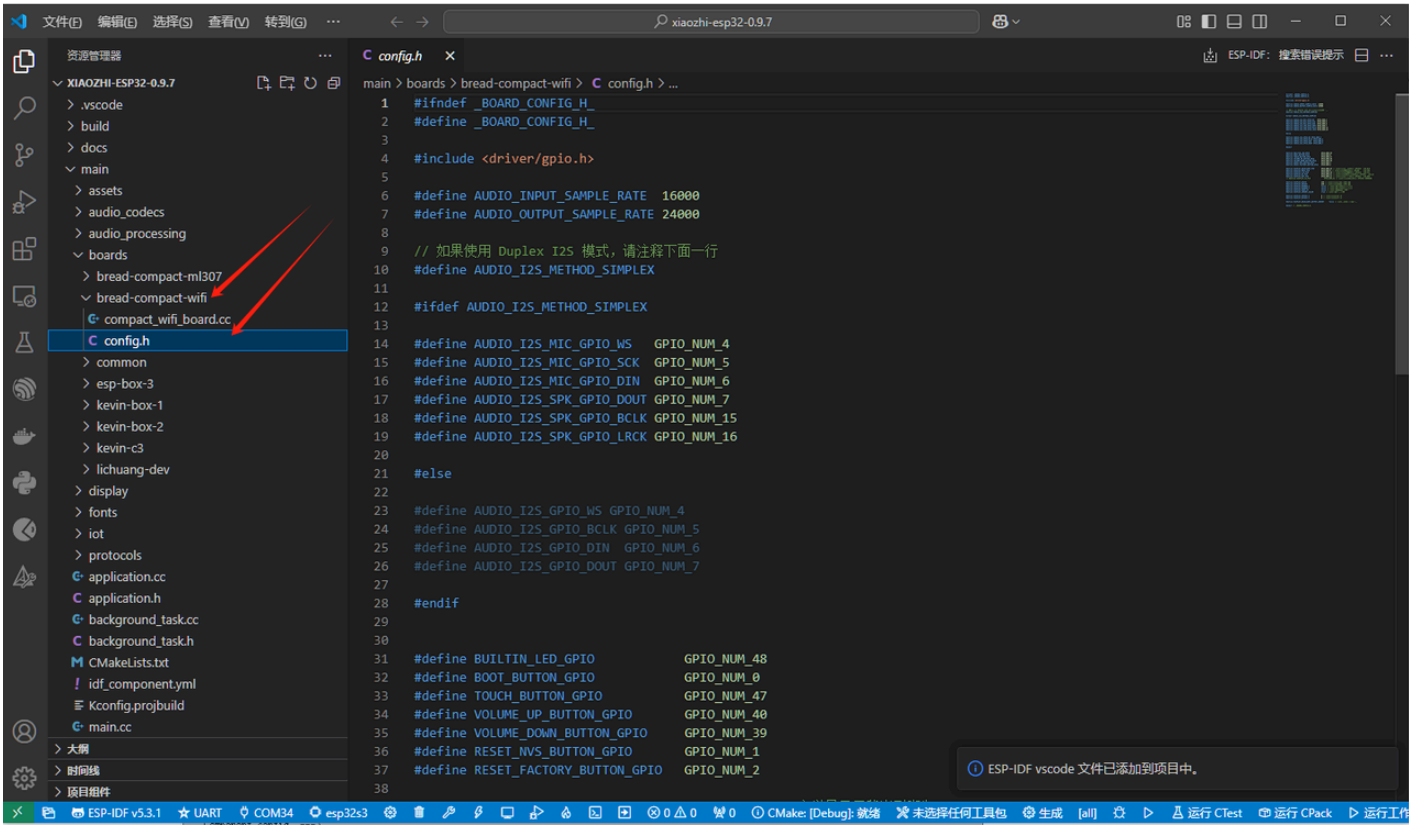
Note: If you clone the code from GitHub, it will not include a version number like 0.9.7 by default. Simply open the directory to \xiaozhi-esp32.



③ After opening the Xiaozhi AI project using VSCode, the interface reference is as shown in the following figure:

The following example has closed the VSCode welcome page and opened the config.h Header File under the xiaozhi-esp32-0.9.7\main\boards\bread-compact-wifi directory, where the pin definitions for the microphone, audio amplifier, etc., as well as the screen setting parameters for the esp32-s3 development board version of the Xiaozhi AI Terminal are all located.

Note: After modifying the source code of Xiaozhi AI, you need to recompile the project with idf.py build, and then compile and flash it to the development board or device using idf.py flash to make it effective.



2. How to modify the pin configuration of the development board?

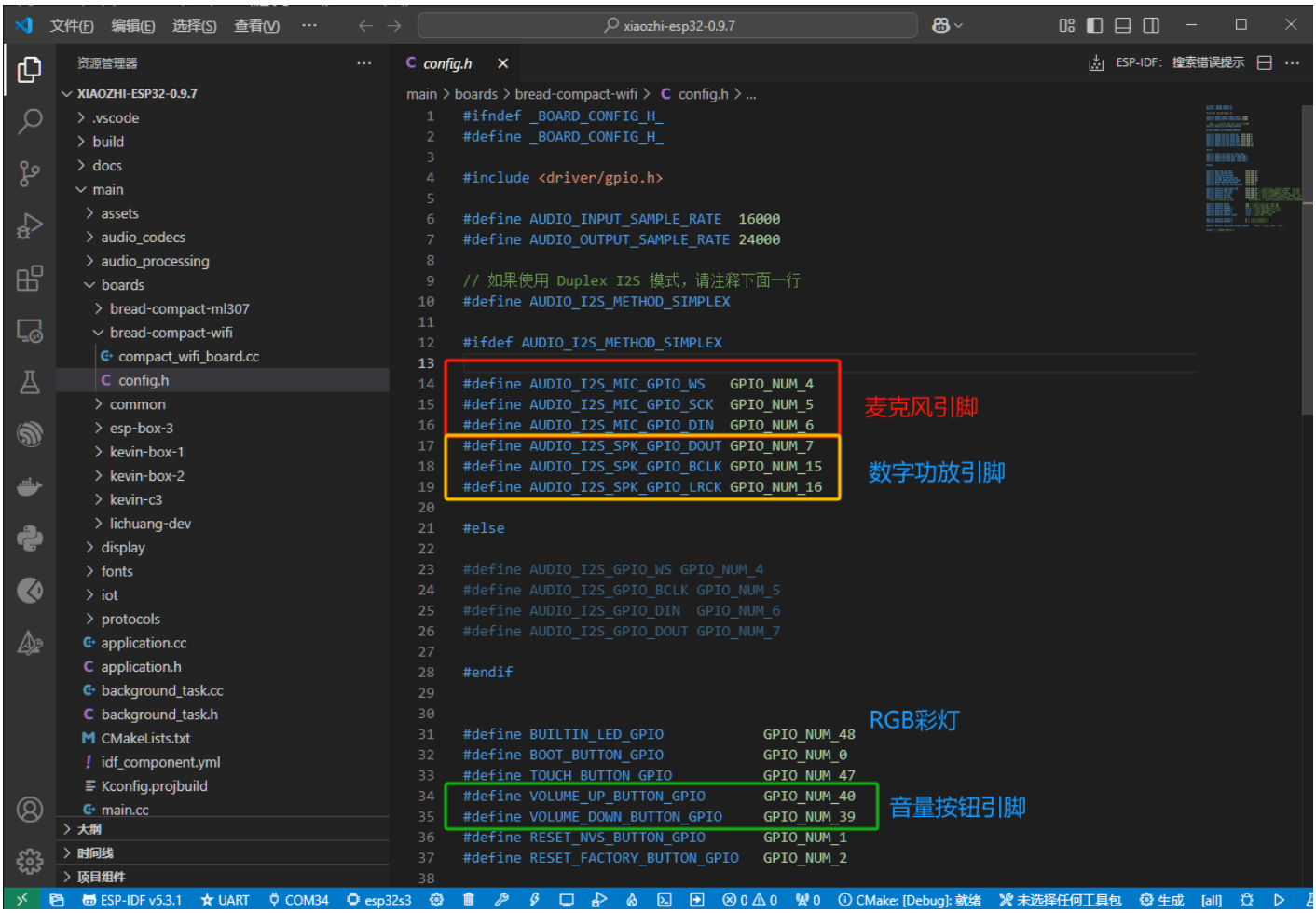
Development boards using the esp32s3 chip module, or self-designed development boards or terminal devices equipped with the esp32s3 chip module, generally modify the code on the breadboard version for adaptation.

1) Source code configuration directory and pin parameters for the breadboard version of the Xiaozhi AI Terminal

The configuration directory for the breadboard wifi version is: xiaozhi-esp32-0.9.7\main\boards\bread-compact-wifi.

The configuration directory for the breadboard ML307-4G version is: xiaozhi-esp32-0.9.7\main\boards\bread-compact-ml307.

Opened the config.h Header File under the xiaozhi-esp32-0.9.7\main\boards\bread-compact-wifi directory, where the pin definitions for the microphone, audio power amplifier, etc., as well as the screen setting parameters for the esp32-s3 development board version of the Xiaozhi AI Terminal are located.



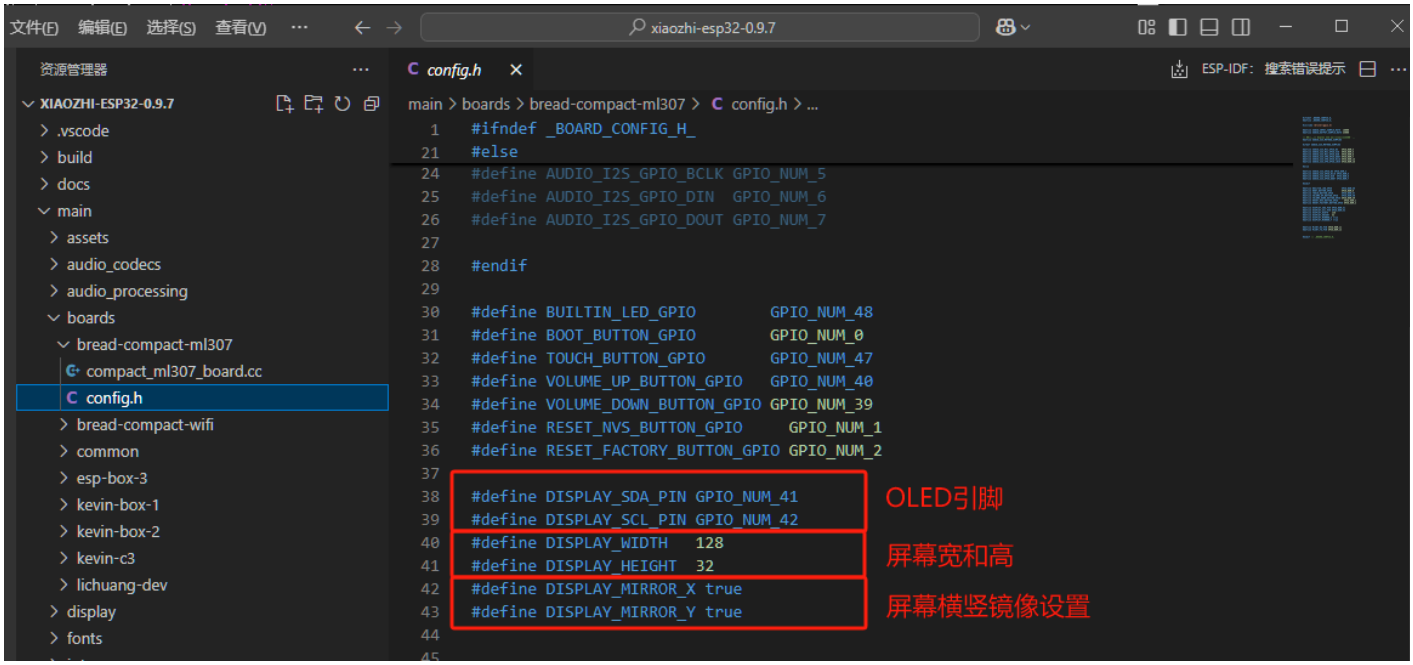
Note: If you need to modify other development boards, please switch to the corresponding development board configuration directory under the \main\boards directory to modify the source code, and at the same time configure the corresponding development board in the menu or SDK. You can refer to the previous document for compiling the Xiaozhi section: Setting up the ESP IDF 5.3 Development Environment on Windows and Compiling Xiaozhi

2)OLED Screen Setting Parameters for Breadboard WIFI Version/ML307R-4G Version

The screen is by default a 0.91-inch OLED screen with an SSD1306 driver chip, with a resolution of 128x32. If you use another 4-wire OLED screen with an SSD1306 driver chip, you need to modify the resolution.

Example 1: If using a 0.96-inch OLED screen with an SSD1306 driver chip and a resolution of 128x64, you need to modify the screen height #define DISPLAY_HEIGHT 64

Example 2: If the OLED screen is upside down by default, the Y-axis needs mirroring settings. For example, modify the mirroring setting parameters: #define DISPLAY_MIRROR_Y false; #define DISPLAY_MIRROR_X false; For more details, please debug on your own.



Please try and verify additional settings on your own.

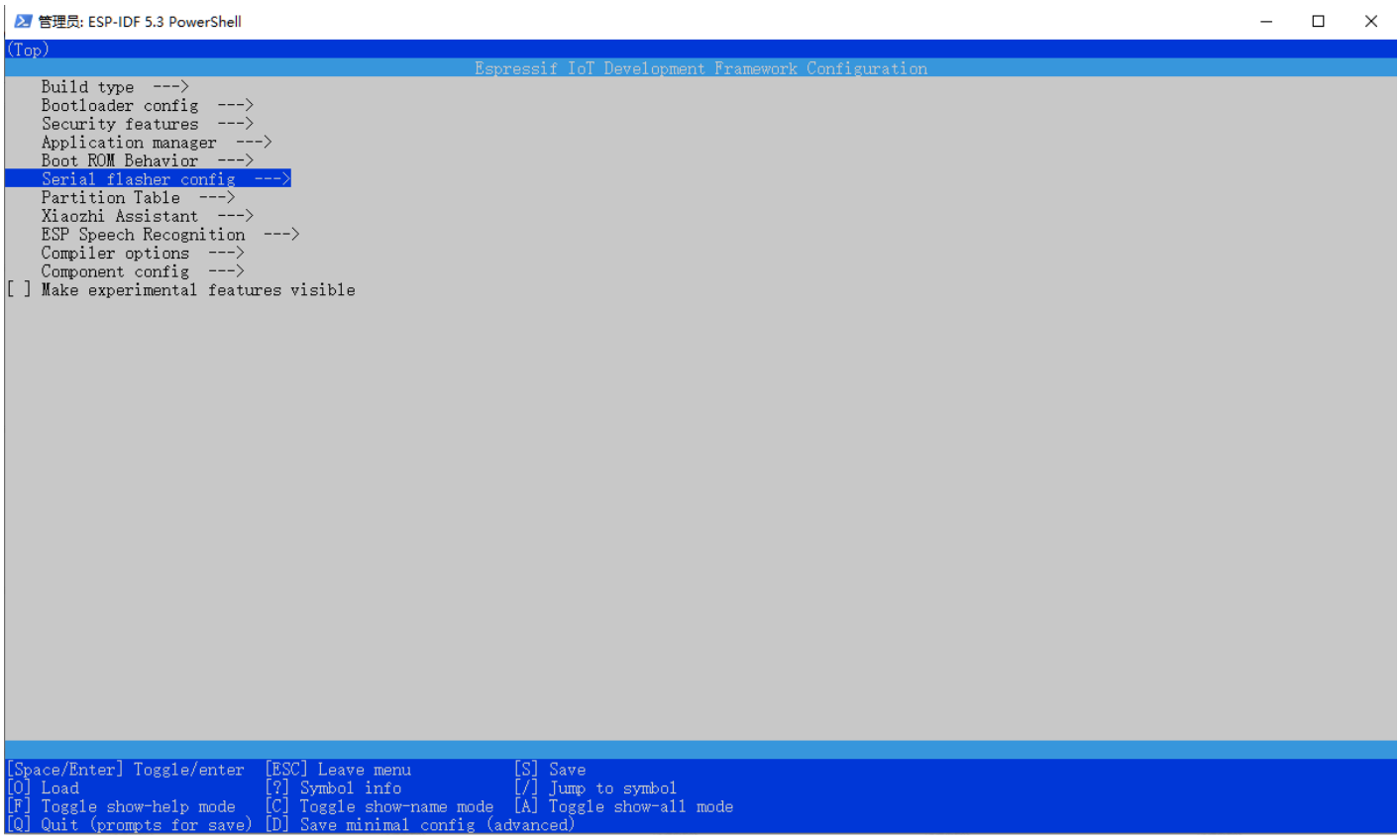
Note: Anymodifications to the Xiaozhi AI source code require recompiling the project using idf.py build, and then using idf.py flash to compile and flash it onto the development board or device to take effect.

3. How to modify the flash size?

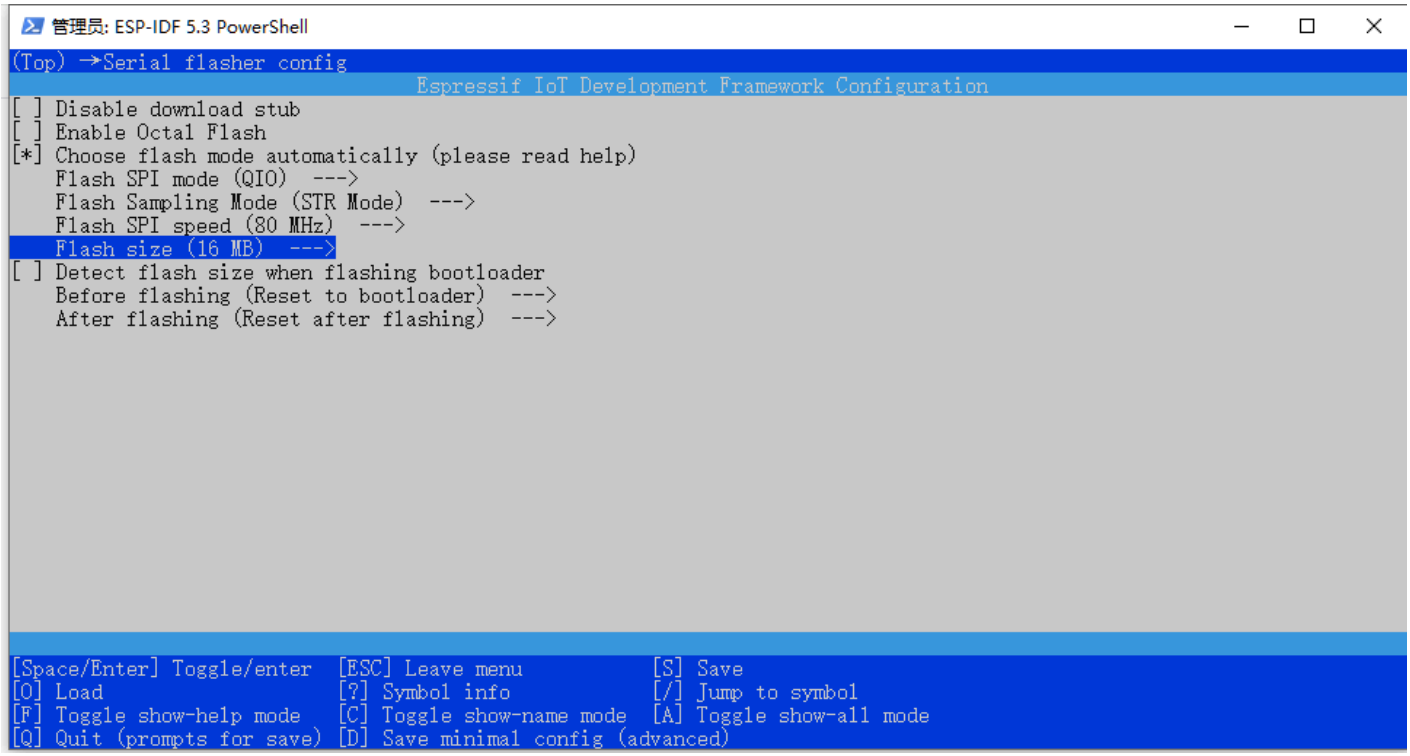
Method 1: Use the IDF 5.3 PowerShell command terminal and the idf.py menuconfig command menu to modify:

Continuing from the previous section, after switching the source code path of Xiaozhi in the ESP IDF 5.3 PowerShell command terminal, please enter the command "idf.py menuconfig" and press Enter. The terminal will load the program and enter the main interface of the IDF menu configuration.

Press the up and down arrow keys on the keyboard to switch options, select the Serial flasher config ---> row menu, and press Enter to enter the configuration, as shown in the figure below:

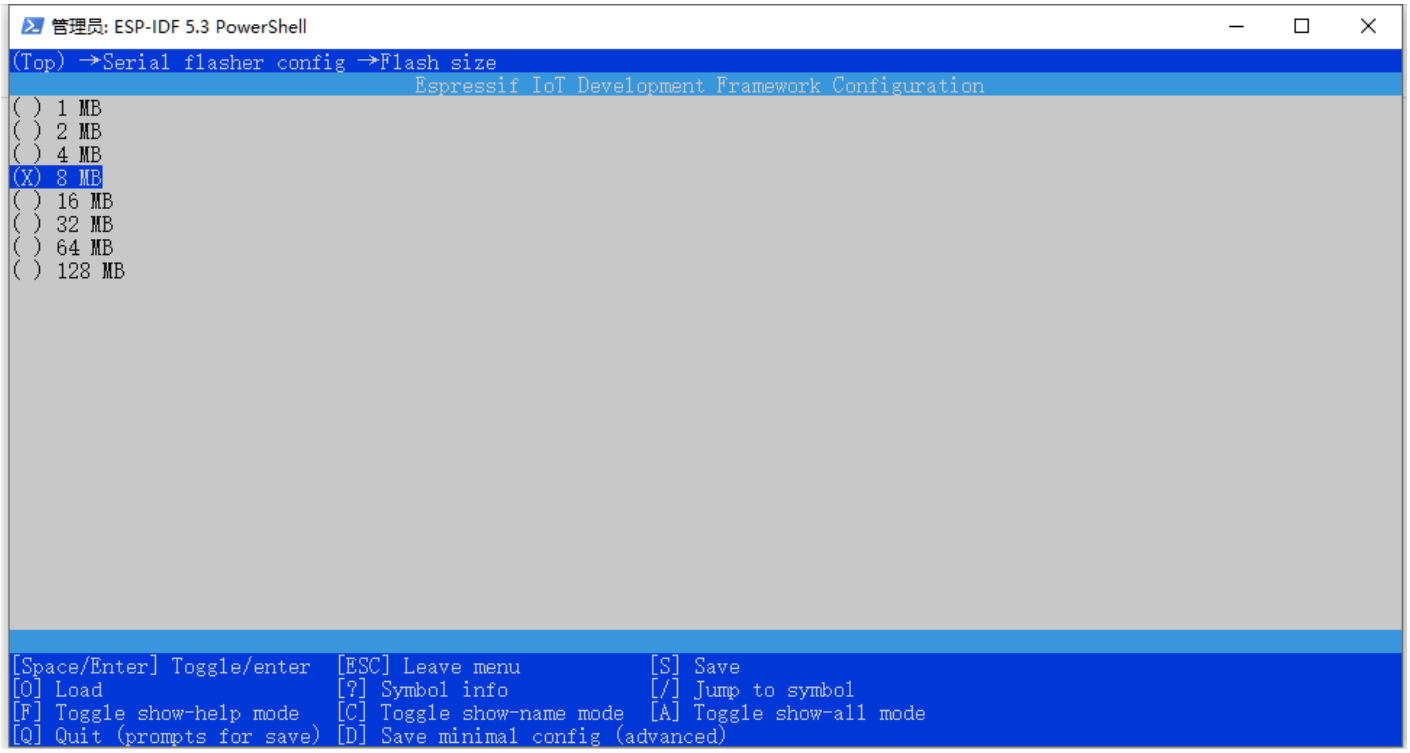


On the Serial flasher config screen, use the up and down arrow keys on the keyboard to switch options and select Flash size (xxMB) --->, then press Enter to enter and modify the flash size. As shown in the figure below:

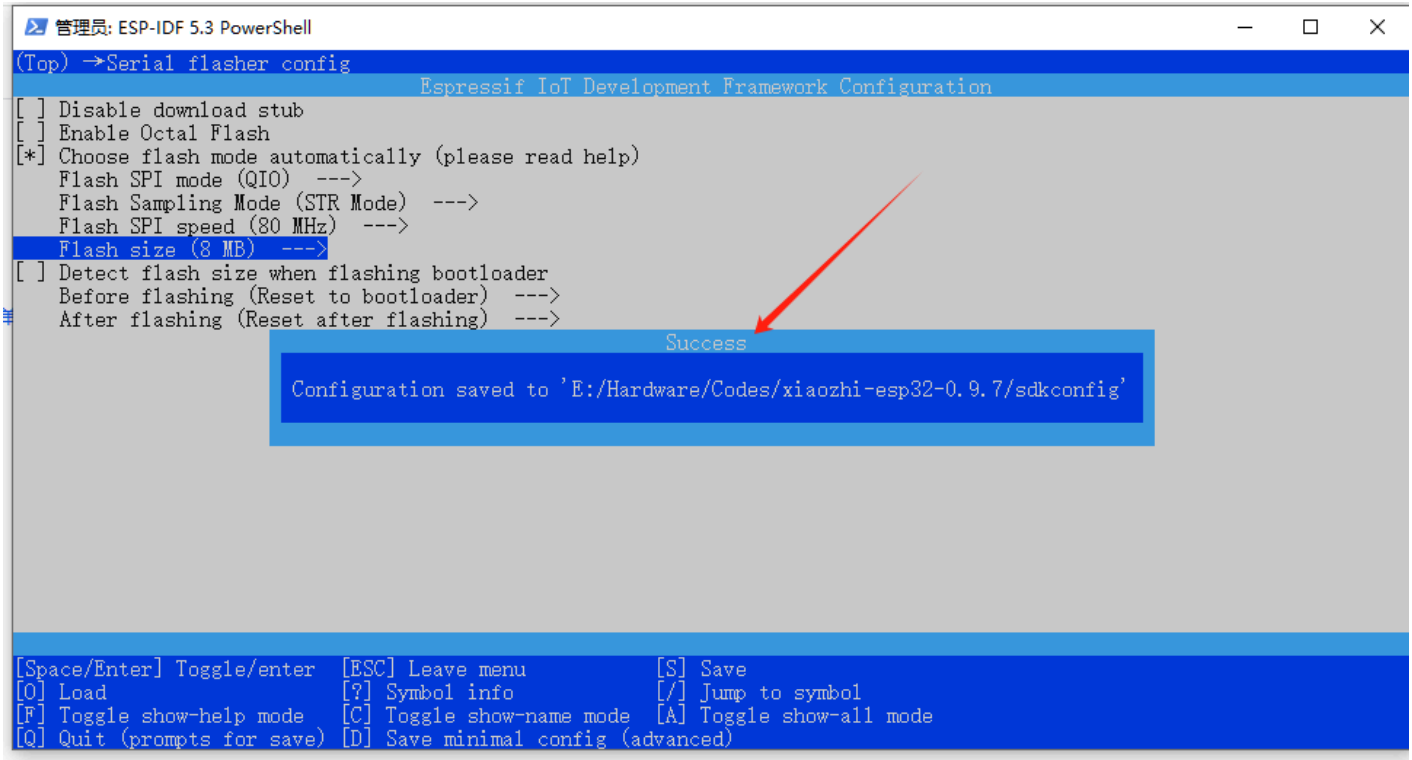


On the Flash size interface, use the up and down arrow keys on the keyboard to switch options, and press Enter to confirm,as shown below:

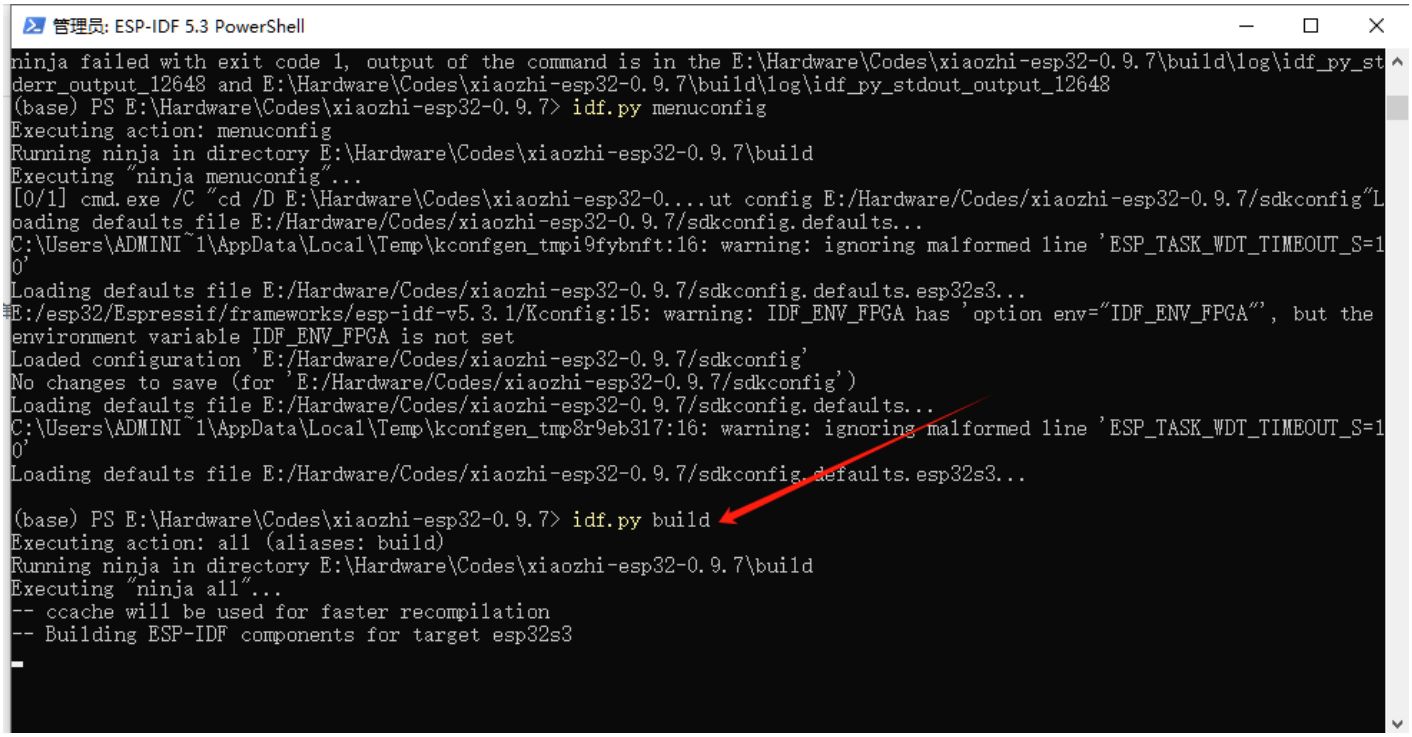
Note: Here we demonstrate modifying the selection to 8M. Please modify it according to the actual flash size of your development board.



When making modifications, you must press the S key to save, wait for the system to prompt Success, then press the Enter key to confirm the closure of the Success prompt, and finally press the Q key or ESC key on the keyboard to exit or leave the menuconfig interface, thus completing the modification operation, as shown in the figure below.



Note: After modification, you need to recompile the project with idf.py build, and the changes will take effect only after the compilation is successful.



Method 2: Modify the SDK configuration using VScode, to be supplemented

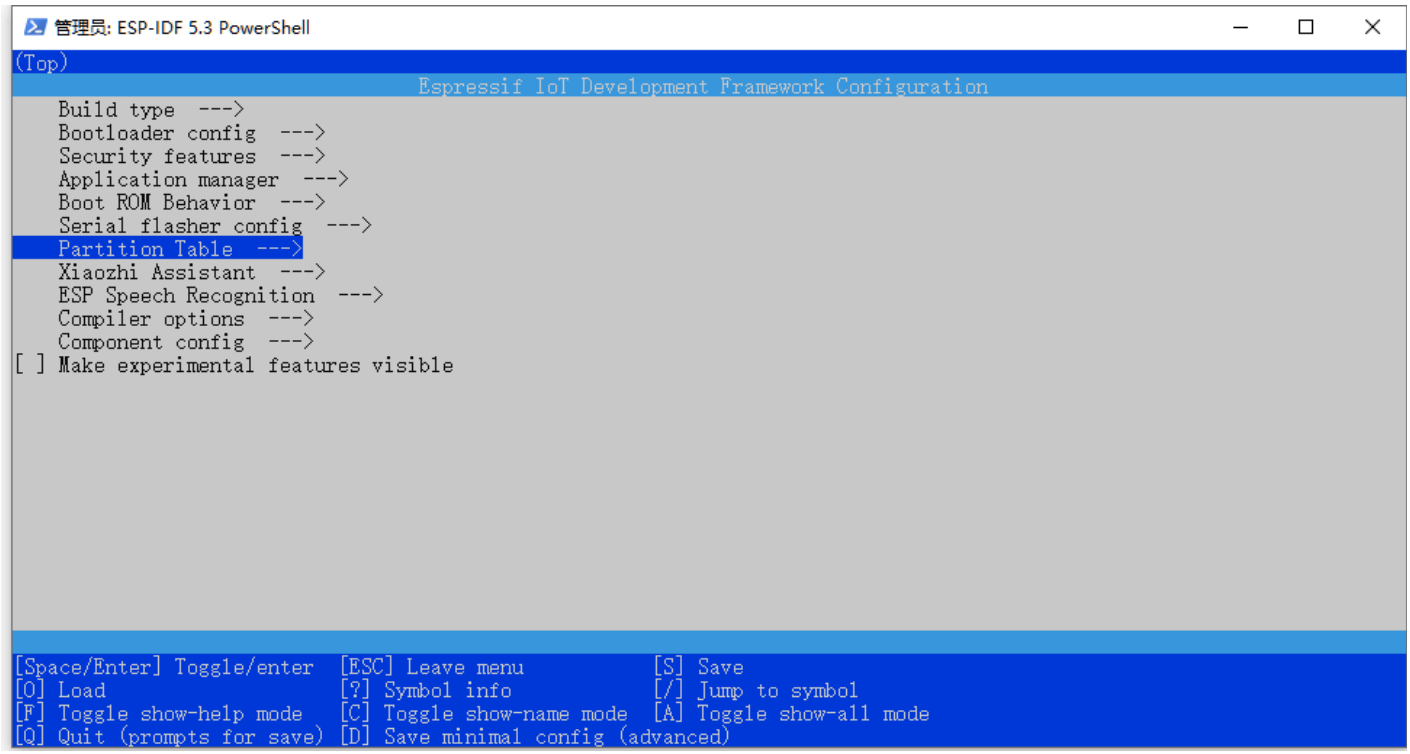
4. How to modify the partition table of the Xiaozhi AI project?

Since the Flash size is not the default 16M or the size of a certain partition needs to be modified, in this case, the partition table in the Xiaozhi AI source code needs to be modified:

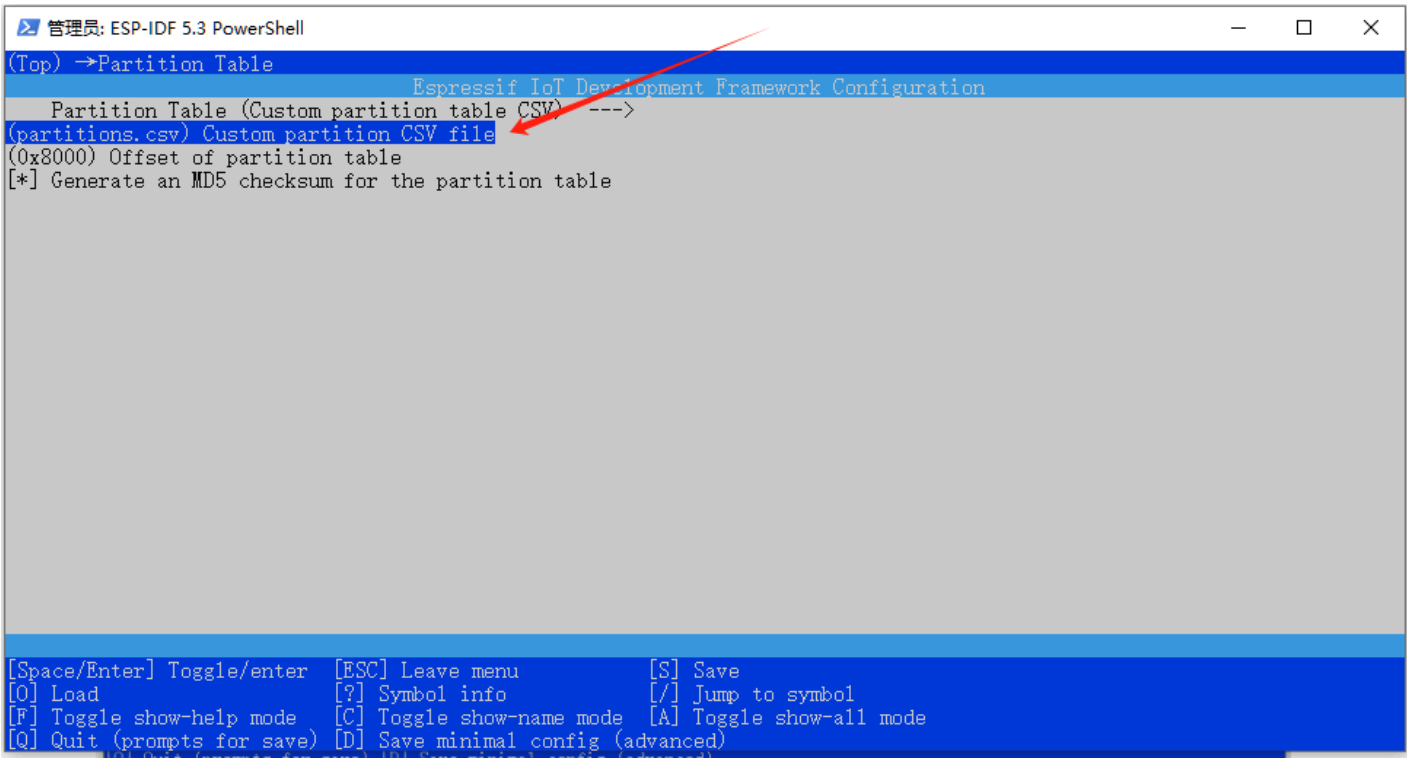
1) (New Version) Modify using the ESP-IDF Powershell Configuration Menu

Note: If the current source code version is greater than 1.0.0, refer to the following steps in this section for modification.

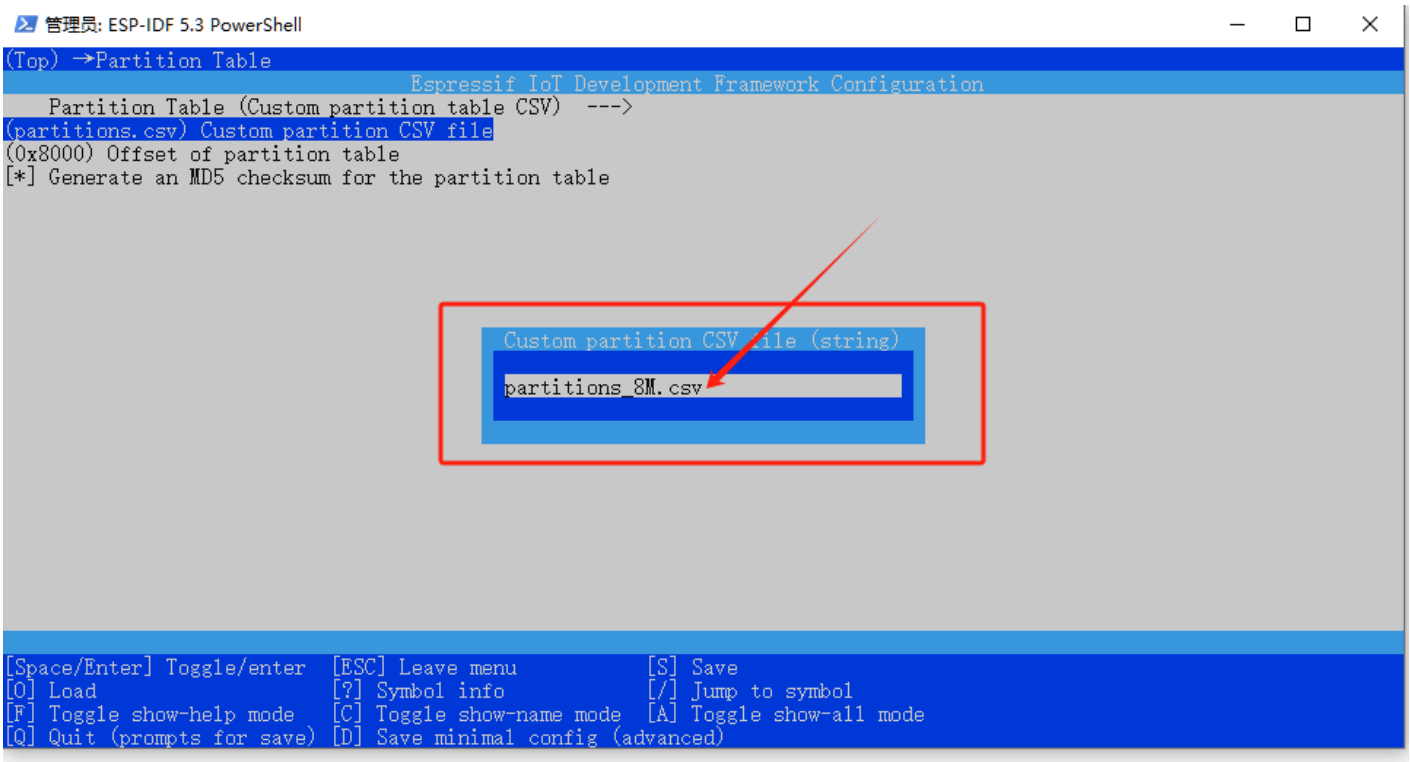
Run ESP-IDF Powershell as an administrator, cd to the root directory of the Xiaozhi AI open-source project, enter idf.py menuconfig to enter the menu configuration interface of the Xiaozhi AI project, and use the up and down buttons to select the Partition Table ----> menu item, as shown in the following figure:



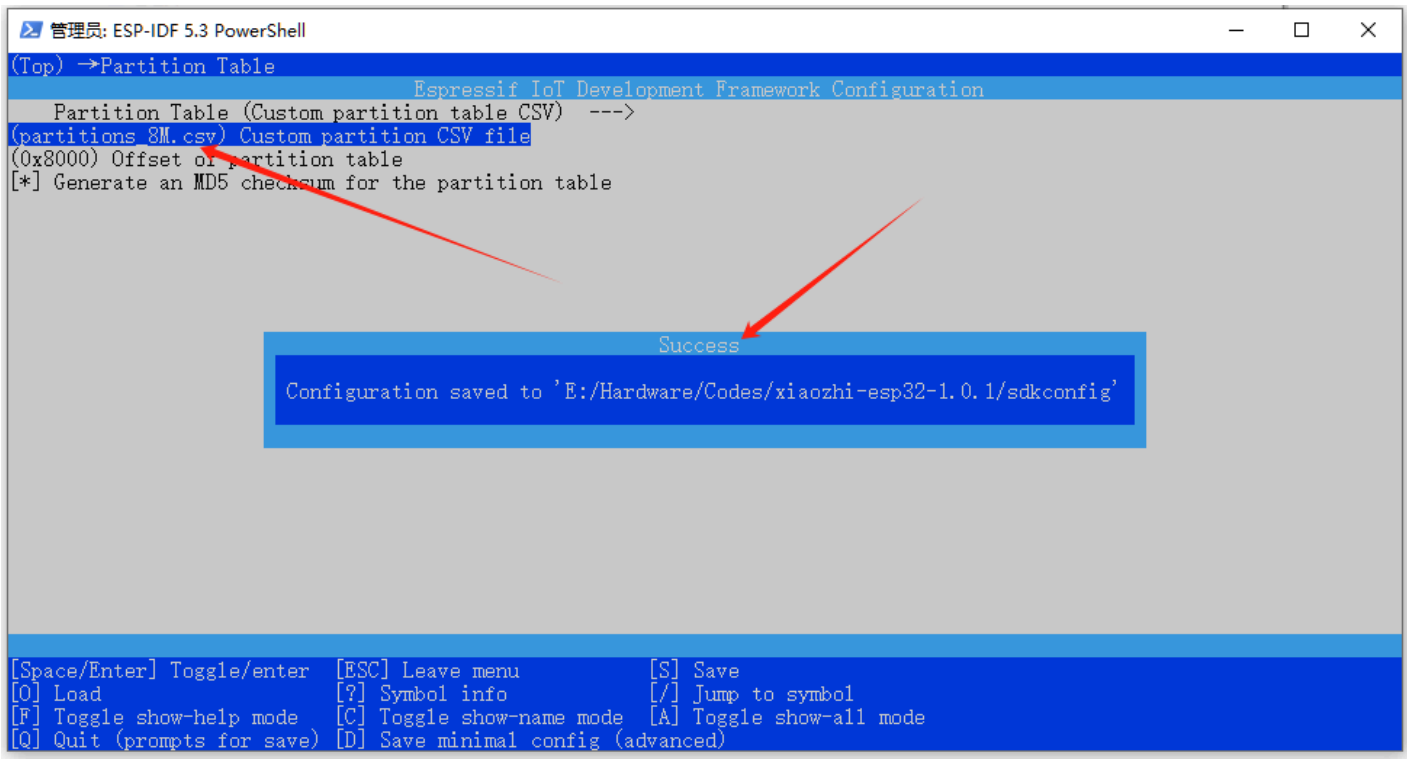
Press the Enter key on the keyboard to enter the Partition Table menu, select the first option in the Partition Table interface (partitions.csv) Custom Partition Table CSV file option (the content in the parentheses may vary if modified previously), and press the Enter key again to enter and modify the custom partition table, as shown in the following figure:



If the user is using an 8M flash, change it to partitions_8M.csv; if the user is using a 4M flash, change it to partitions_4M.csv. The Xiaozhi open source project already includes these two partition table CSV files. Modifying the 8M flash is shown in the following figure:



Press the 【S】 key, then press the Enter key to confirm and save, and then press 【Q】 to exit.



Note: After modifying the partition table configuration, you need to save it and then recompile the project for the changes to take effect.

Meanwhile, please ask the user to confirm that the flash size setting is consistent with the partition table (refer to Section 3 for modifying the flash size).

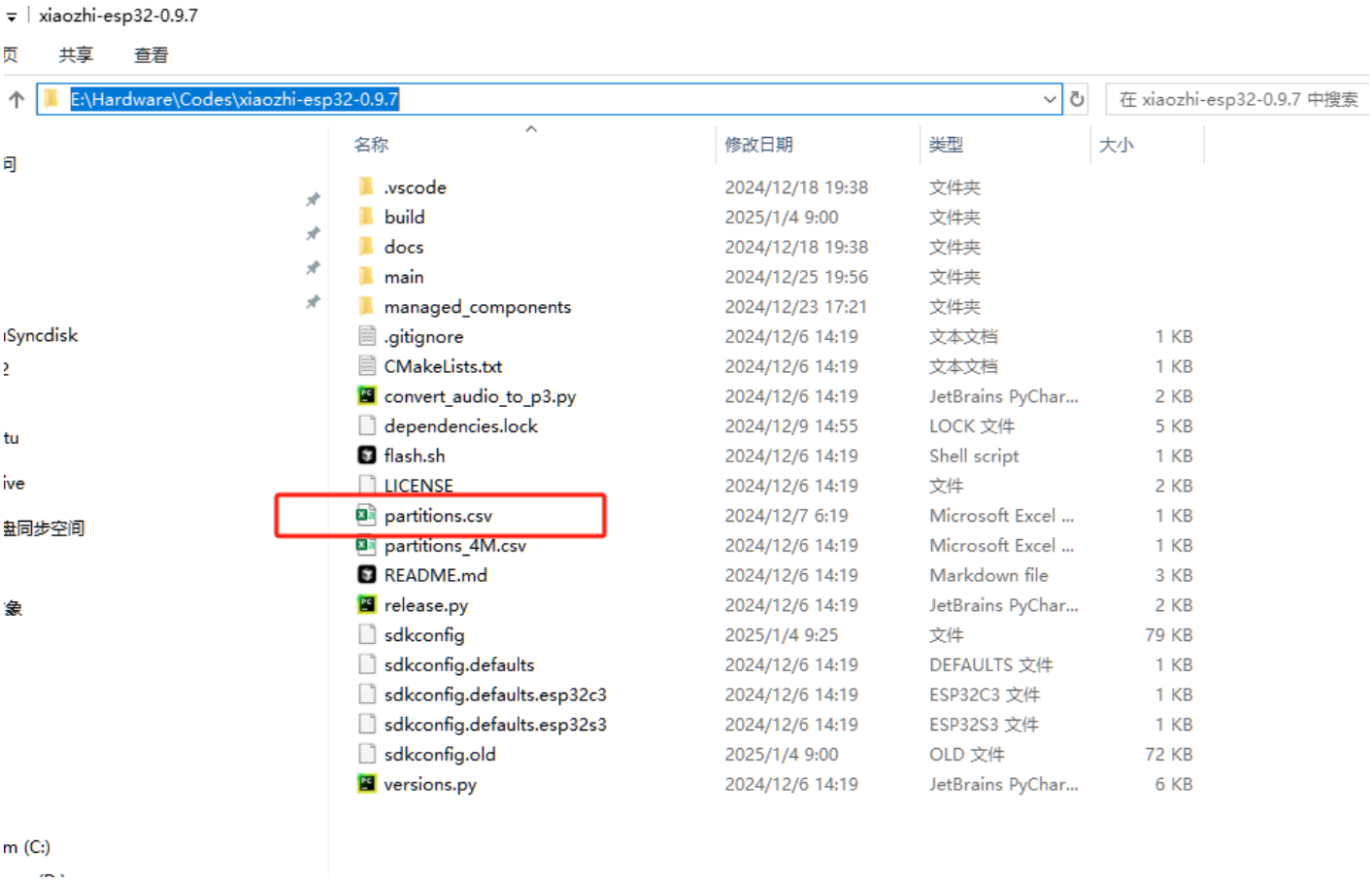
2) (Old Version) Compile after directly modifying the partition table file

Note: If the current source code version is less than 1.0.0, you can refer to the instructions in this section for modification.

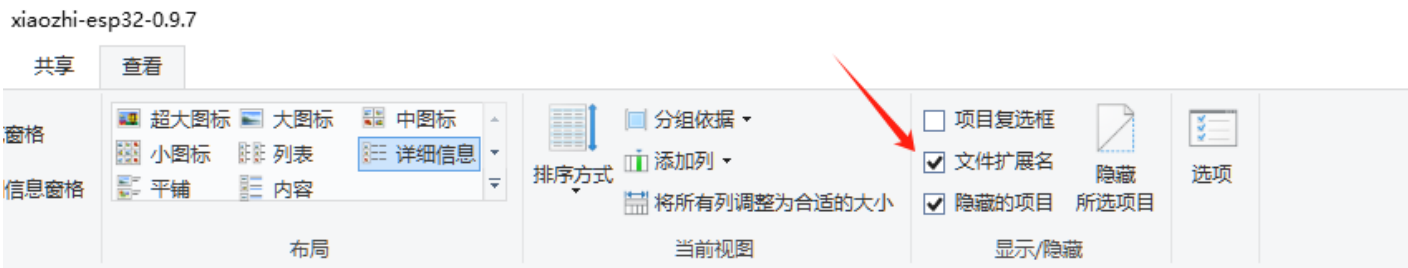
Open the root directory of the source code where the Xiaozhi AI project is stored, and it is recommended to open the partitions.csv file with a text editor.

Note: The path shown in the figure below is for reference only. Please open it to the root directory based on the actual location of the source code on your own computer.

The partitions.csv file is opened by Excel by default. It is recommended to open it with a text editor. If the file extension is not displayed, please go to



If the file partitions.csv does not show its file extension, please open it in the browser's view, as shown in the following figure:



(1)Explanation of partitions.csv partition table

(1) The size set in the partitions.csv partition table cannot exceed the flash size of the ESP32 chip on the development board, which generally corresponds to the number after N in the development board model. For example, the number after N in the ESP32-S3-N16R8 chip module is 16, indicating a flash size of 16M.

(2) The partitions.csv partition table is generally modified according to the needs of the development program application, referring to the official recommended examples and specifications. You only need to modify the Offset and Size columns. Offset can be left unconfigured, and the system will automatically calculate the offset position based on Size. The Type and SubType qualifiers are supported (refer to the official Espressif documentation for details), so please do not modify them casually, otherwise compilation errors will occur.

(2)**Reference example for modifying the partitions.csv partition table file**

- **Example 1: Flash size of 8M corresponds to modifying partitions.csv**

Since the flash size is less than 16M, it is recommended to modify the file by removing the two lines of configuration for ota_0 and ota_1, with the remaining Configuration Items totaling 8M.

Note: After removing the OTA configuration line, OTA automatic upgrade will no longer be available. If the program cannot be automatically upgraded, manual flashing will be required. Meanwhile, it is recommended to turn off the automatic upgrade setting in the website backend to avoid potential errors that may occur when the running program detects a new version but fails to download and upgrade.

Note: Currently, the merged bin firmware of the Xiaozhi AI Terminal esp32s3 requires approximately 4M in size. For the configuration lines of factory, app, etc. in the partitions.csv partition table, it is recommended to set them to 4M, and do not modify the configuration size; otherwise, the flashing process may not complete successfully.

Note: After modifying the partitions.csv partition table, you need to recompile the project. The changes will take effect only after idf.py build succeeds.

More operation instructions to be supplemented. If you have any questions, please discuss them in the QQ group. Thank you!